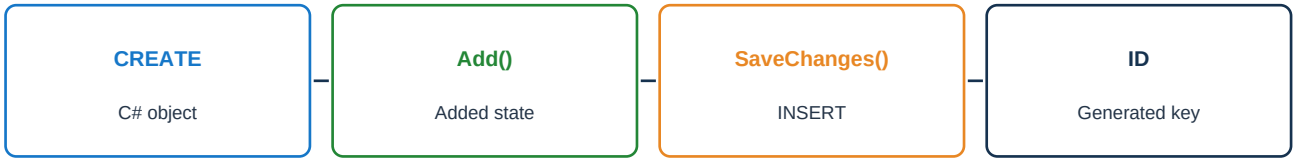
Insert & Delete in EF Core

Visual Lesson Summary • Change Tracker, generated IDs, and SQL logging

1. Core Idea

EF Core prepares database commands based on entity states. **Add()** marks an entity as **Added**; **Remove()** marks a tracked entity as **Deleted**. **SaveChanges()** executes the database operation.

2. Insert Workflow



Insert a new student and retrieve the generated ID

```
var newStudent = new Student
{
        FirstName = "Ali",
        LastName = "Hassan",
        Email = "ali.hassan@example.com",
        Status = "Active",
        RegisteredAt = DateTime.Now
};

context.Students.Add(newStudent);
int affectedRows = context.SaveChanges();

int newStudentId = newStudent.StudentId;
```

After SaveChanges() succeeds, the database-generated key is available on the entity (assuming StudentId is configured as a generated key).

3. Delete Workflow

```
var student = context.Students
        .FirstOrDefault(s => s.StudentId == studentId);

if (student == null)
        return;

context.Students.Remove(student);
int affectedRows = context.SaveChanges();
```

The query loads a tracked entity. Remove() marks it Deleted; SaveChanges() executes the DELETE command.

4. ToQueryString() vs Runtime Logging

Operation	How to inspect SQL	What it shows
SELECT	**ToQueryString()**	Previews SQL for an IQueryable query before execution.
INSERT / DELETE	**Runtime logging**	Shows commands actually executed during SaveChanges().

Why not ToQueryString() for INSERT or DELETE?

ToQueryString() works with **IQueryable** queries, such as SELECT. INSERT and DELETE commands are produced and executed by **SaveChanges()**, so inspect them through runtime logging.

```
.LogTo(Console.WriteLine, LogLevel.Information)
```

Runtime logging prints SQL commands as EF Core executes them. Sensitive-data logging can expose parameter values, so use it carefully.

5. Important Sequences

Insert: Create object → Add() → SaveChanges() → read generated ID.
Delete: Query row → check for null → Remove() → SaveChanges().

6. Mental Anchor

> **EF Core builds the command from the Change Tracker state — SaveChanges() executes it.**